3.5.2 Physical, Physiological and Psychological changes

Body composition

Body composition changes during the period of maturation. The changes occur due to hormonal influences which regulate the development of sex characteristics. The skeletal growth continues for a longer period of time for boys than girls. Usually the skeleton reaches its full maturity by 17 years for girls and by 20 years for boys. As mineralization increases the water content decreases.

Girls tend to deposit more fat whereas boys add more muscle mass. The result of pubertal changes is that boys have more lean body mass, skeletal weight and less adipose tissue as a ratio of total body mass. This difference in body composition for boys and girls is reflected in their nutrient requirements.

Sexual maturity

The growth spurt is accompanied by sexual maturity. In girls there is development of breasts, auxiliary and pubic hair and menarche.

In boys the pubertal changes include deepening of voice, broadening of shoulders, development of auxiliary and pubic hair, growth of penis and testicles.

Psycho social changes

As this period is a transition to adulthood, they try to develop self-identity. The desire to be accepted in their peer group changes their food habits, dressing and group conduct. This in turn brings psychological, emotional and social stress.

3.5.3 Nutritional requirements

Adequate intake of nutrients during the adolescent period is critical in the phenotype expression of genetic potential of bone tissue growth and mineralization. Insufficient intake of nutrients inhibits growth promoting hormones, impeding or delaying the onset of pubertal development, causes poor concentration on studies and may even lower height gain during this period.

The Recommended Dietary Allowance for adolescents is presented in Table 3.3

Table. 3.3 ICMR recommended dietary allowances for Adolescents

Nutrient	Years			
	13-15		16-17	
	Boys	Girls	Boys	Girls
Weight (Kg)	47.6	46.6	55.4	52.1
Energy (kcal)	2750	2330	3020	2440
Protein (g)	54.3	51.9	61.5	55.5
Fat (g)	45	40	50	35
Calcium (mg)	800	800	800	800
Iron (mg)	32	27	28	26
Vitamin A (µg)	600	600	600	600
Beta carotene (µg)	4800	4800	4800	4800
Thiamine (mg)	1.4	1.2	1.5	1.07
Riboflavin (mg)	1.6	1.4	1.8	1.2
Nicotinic acid (mg)	16	14	17	14
Pyridoxine (mg)	2	2	2	2
Ascorbic acid (mg)	40	40	40	40
Folic acid (µg)	150	150	200	200
Vitamin B12(µg)	0.2-1	0.2-1	0.2-1	0.2-1

Source: Dietary Guidelines for Indians - A Manual by Kamala Krishnaswamy, B. Sesikeran (Second Edition 2011) , NIN, ICMR

Energy

The metabolic demands of growth and energy expenditure increases the calorie needs. The ICMR committee has suggested that energy should be provided on the basis of ideal weight for age. The energy requirements for boys are more than that of girls which is 2750 kcal and 3020 kcal for boys 13-15 years and 16-18 years respectively and 2330 kcal

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